THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Easy level: direct, single-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Which of the following pairs (x, y) is a solution of $3x + y = 11$? 1

(a) $(5, 2)$ **(b)** $(2, 5)$ **(c)** $(-2, 5)$ **(d)** $(2, -5)$

2. Which of the following is a root of $x^2 - 5x + 6 = 0$? 1

(a) 3 **(b)** 4 **(c)** 7 **(d)** -3

3. The 8th term of the AP $5, 8, 11, 14, \ldots$ is: 1

(a) 29 **(b)** 23 **(c)** 26 **(d)** 24

4. The ratio of corresponding sides of two similar triangles is 2:5. Their perimeters are in the ratio 2: 1

(a) 2 **(b)** 5 **(c)** 7 **(d)** 10

5. The midpoint of the join of $(0,\ 0)$ and $(8,\ 6)$ is: 1

(a) $(4,\ 3)$ **(b)** $(5,\ 3)$ **(c)** $(8,\ 6)$ **(d)** $(4,\ 4)$

6. In a right triangle with perpendicular 12, base 5 and hypotenuse 13, $\tan\theta =$ 1

(a) $\frac{5}{13}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{12}{13}$ **(d)** $\frac{5}{12}$

7. From a point on level ground, 12 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. The height of the tower is: 1

(a) 12 m **(b)** $12\sqrt{3}$ m **(c)** 24 m **(d)** $4\sqrt{3}$ m

8. A point P is 10 cm from the centre O of a circle of radius 6 cm. The length of the tangent drawn from P to the circle is: 1

(a) 64 cm **(b)** 16 cm **(c)** 8 cm **(d)** 4 cm

9. A quadrant of a circle has radius 7 cm. The perimeter of the quadrant is: 1

(a) 11 cm **(b)** 18 cm **(c)** 25 cm **(d)** 36 cm

10. A right circular cylinder has base radius 14 cm and height 10 cm. The curved surface area of the cylinder is: 1

(a) 2112 cm² **(b)** 880 cm² **(c)** 6160 cm² **(d)** 1760 cm²

11. For the data with classes 0–10, 10–20, 20–30, 30–40 with frequencies 3, 9, 5, 8 respectively, The median of the data is: 1

(a) 30 **(b)** 20 **(c)** 22.2 **(d)** 21

12. A die is thrown once. The probability of getting an odd number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{3}{7}$

13. The number of prime factors (counted with repetition) of 30 is: 1

(a) 8 **(b)** 3 **(c)** 2 **(d)** 4

14. The sum of the zeroes of $x^2 - 4x - 5$ is: 1

(a) 4 **(b)** -5 **(c)** -1 **(d)** -4

15. If $x = 4, y = 2$ is a solution of $3x + y = k$, then k is: 1

(a) 24 **(b)** 14 **(c)** 10 **(d)** 18

16. In $3x^2 + 7x - 2 = 0$, the value of b is: 1

(a) 3 **(b)** -7 **(c)** -2 **(d)** 7

17. The sum of the first 8 terms of $2, 4, 6, 8, \ldots$ is: 1

(a) 90 **(b)** 144 **(c)** 56 **(d)** 72

18. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$, $DB = 4$, $AE = 6$. Then $EC =$ 1

(a) 6 **(b)** 4 **(c)** 8 **(d)** 12

19. The point $(-2, 5)$ lies in the: 1

(a) first quadrant **(b)** third quadrant **(c)** fourth quadrant **(d)** second quadrant

20. The value of $\cos 60^\circ$ is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** $\frac{1}{2}$

21. From a point on level ground, 8 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $8\sqrt{3}$ m **(b)** $16\sqrt{3}$ m **(c)** 8 m **(d)** $\frac{8\sqrt{3}}{3}$ m

22. The length of the tangent from an external point P to a circle is 12 cm and P is 13 cm from the centre. The radius of the circle is: 1

(a) 25 cm **(b)** 1 cm **(c)** 5 cm **(d)** 13 cm

23. The radius of a circle is 14 cm. The area of the circle is: 1

(a) 1232 cm² **(b)** 616 cm² **(c)** 44 cm² **(d)** 88 cm²

24. A right circular cylinder has base radius 14 cm and height 5 cm. The volume of the cylinder is: 1

(a) 1672 cm³ **(b)** 3080 cm³ **(c)** 6160 cm³ **(d)** 440 cm³

25. The upper limit of a class 20–30 is: 1

(a) 30 **(b)** 20 **(c)** 25 **(d)** 10

26. A die is thrown once. The probability of getting a perfect square is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{6}$

27. Which of the following is an irrational number? 1

(a) 36 **(b)** $\frac{6}{9}$ **(c)** $\sqrt{11}$ **(d)** $\sqrt{36}$

28. The product of the zeroes of $x^2 - 4x - 5$ is: 1

(a) 5 **(b)** -1 **(c)** 4 **(d)** -5

29. The sum of two numbers is 14 and their difference is 6. The larger number is: 1

(a) 8 **(b)** 10 **(c)** 20 **(d)** 4

MASTERMIND

30. The roots of $(x+1)(x-4)=0$ are: 1

(a) 1 and -4 **(b)** -1 and -4 **(c)** -1 and 4 **(d)** 3 and -4

31. Which of the following sequences is an AP? 1

(a) $9, 7, 6, 5, \ldots$ **(b)** $9, 7, 5, 3, \ldots$ **(c)** $9, 7, 3, 1, \ldots$ **(d)** $9, 7, 3, -3, \ldots$

32. The areas of two similar triangles are in the ratio 9:25. The ratio of their corresponding sides is 3: 1

(a) 15 **(b)** 8 **(c)** 25 **(d)** 5

33. The coordinates of the point on the x-axis at distance 4 from the origin on the left are: 1

(a) $(4,\ 0)$ **(b)** $(0,\ -4)$ **(c)** $(-4,\ -4)$ **(d)** $(-4,\ 0)$

34. The value of $\sin^2 35^\circ + \cos^2 35^\circ$ is: 1

(a) 35 **(b)** 2 **(c)** 0 **(d)** 1

35. From a point on level ground, 12 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) 4 m **(b)** 12 m **(c)** $12\sqrt{3}$ m **(d)** $4\sqrt{3}$ m

36. A tangent to a circle of radius 15 cm has length 20 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 50 cm **(b)** 25 cm **(c)** 5 cm **(d)** 35 cm

37. The radius of a circle is 14 cm. The circumference of the circle is: 1

(a) 44 cm **(b)** 88 cm **(c)** 176 cm **(d)** 616 cm

38. A right circular cone has base radius 14 cm and height 48 cm. The slant height of the cone is: 1

(a) 50 cm **(b)** 31 cm **(c)** 47 cm **(d)** 62 cm

39. The class interval is 20–30. The class mark is: 1

(a) 30 **(b)** 25 **(c)** 10 **(d)** 20

40. A coin is tossed two times. The probability of getting at least one head is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{3}{4}$ **(c)** 1 **(d)** $\frac{1}{2}$

41. The prime factorisation of 84 is: 1

(a) $2^3 \times 11$ **(b)** 2×41 **(c)** $2^2 \times 3 \times 7$ **(d)** 2×43

42. A polynomial of degree 3 has at most how many zeroes? 1

(a) infinitely many **(b)** 4 **(c)** 3 **(d)** 2

43. The graph of $x = 3$ is a line parallel to the: 1

(a) x-axis **(b)** y-axis **(c)** line $y = x$ **(d)** none of the axes

44. The discriminant of $2x^2 + 3x + 1 = 0$ is: 1

(a) 17 **(b)** 1 **(c)** -1 **(d)** 5

45. In an AP with $a = 8$ and $d = -2$, the 2nd term is: 1

(a) 4 **(b)** 10 **(c)** 6 **(d)** -16

46. The diagonal of a rectangle with sides 5 and 12 is: 1

(a) 17 **(b)** 13 **(c)** 169 **(d)** 7

47. The distance between the points $(5, 0)$ and $(0, 12)$ is: 1

(a) 13 **(b)** 17 **(c)** 169 **(d)** 7

48. $\sin 40^\circ = \cos$ of: 1

(a) 55° **(b)** 140° **(c)** 40° **(d)** 50°

49. A vertical pole is 15 m high. The angle of elevation of the sun is 45° and the pole casts a shadow on level ground. The length of the shadow is: 1

(a) $\frac{15}{2}$ m **(b)** $15\sqrt{3}$ m **(c)** 30 m **(d)** 15 m

50. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 6 cm. The length of each tangent is 8 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 28 cm **(b)** 14 cm **(c)** 32 cm **(d)** 16 cm

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND